MATLAB Project

Let $f(x)$ be a function of period 2π such that $f(x) = x^2$ in the range $-\pi \le x \le \pi$. The Fourier series for $f(x)$ is defined as

$$\hat{f}(x) = \frac{\pi^2}{3} - 4\sum_{n=1}^{\infty}(-1)^{n+1}\frac{\cos(nx)}{n^2}.$$

Compute the following :

1. Using MATLAB subplot to sketch Fourier series expansion for twenty terms (n from 1 to 20) in one figure and resulting signal in another figure.
2. Plot the line spectrum of the Fourier series expansion given in 1.
3. Estimate the number of Fourier terms (n) such that the Mean Square error (MSE) is equal or less than 0.001. MSE is defined as below

$$MSE = \frac{1}{T}\sum_{i=1}^{T}\left(f(x_i) - \hat{f}(x_i)\right)^2 \le 0.001,$$

where T is total number of samples over $-\pi \le x \le \pi$.